FORMULAS

Formulas are mathematical "recipes" that relate various quantities or terms, so that if the value of one is unknown, it can be worked out if the values of the other terms in the formula are known.

Interest

There are two types of interest – simple and compound. In simple interest, the interest is paid only on the capital. In compound interest, the interest itself earns interest.

amount saved (capital)
interest rate
number of years

$$\text{Interest} = P \times R \times T$$

total interest after T years

◁ **Simple interest formula**
To find the simple interest made after a given number of years, substitute real values into this formula.

amount saved
interest rate
number of years

$$\text{Amount} = P\left(1 + R\right)^{T}$$

total value of investment after T years

◁ **Compound interest formula**
To find the total value of an investment (capital + interest) after a given number of years, substitute values into this formula.

Formulas in algebra

Algebra is the branch of mathematics that uses symbols to represent numbers and the relationship between them. Useful formulas are the standard formula of a quadratic equation and the formula for solving it.

squared value of x multiplied by a number
constant number with no x terms

$$ax^2 + bx + c = 0$$

x multiplied by a number

△ **Quadratic equation**
Quadratic equations take the form shown above. They can be solved by using the quadratic formula.

this means add or subtract

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

△ **The quadratic formula**
This formula can be used to solve any quadratic equation. There are always two solutions.

symbol for pi

$$\pi = 3.14$$

value to 2 decimal places

value to 20 decimal places

$$3.14159265358979323846$$

◁ **The value of pi**
Pi occurs in many formulas, such as the formula used for working out the area of a circle. The numbers after the decimal point in pi go on for ever and do not follow any pattern.

Formulas in trigonometry

Three of the most useful formulas in trigonometry are those to find out the unknown angles of a right triangle when two of its sides are known.

$$\sin A = \frac{\text{opposite}}{\text{hypotenuse}}$$

△ **The sine formula**
This formula is used to find the size of angle A when the side opposite the angle and the hypotenuse are known.

$$\cos A = \frac{\text{adjacent}}{\text{hypotenuse}}$$

△ **The cosine formula**
This formula is used to find the size of angle A when the side adjacent to the angle and the hypotenuse are known.

$$\tan A = \frac{\text{opposite}}{\text{adjacent}}$$

△ **The tangent formula**
This formula is used to find the size of angle A when the sides opposite and adjacent to the angle are known.